

Ahmed Emad | Back-End Engineer | Node.js

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PROFESSIONAL SUMMARY

Node.js Developer focused on building scalable and high-performance server-side applications. Experienced in architecting RESTful APIs, database management (SQL/NoSQL), and deploying robust services. Utilizes strong foundational skills in JavaScript, Next.js, and TypeScript to deliver comprehensive full-stack solutions. Committed to writing clean, maintainable code and continuously learning emerging technologies.

EDUCATION

Bachelor of Information Systems

Faculty of Computers and Artificial Intelligence, Fayoum University, Faiyum | 2022 – 2026 (Expected)

- Major: Information Systems | Minor: Computer Science

TECHNICAL SKILLS

Back-End Development:

- Runtime: Node.js (Understanding of its non-blocking nature and Event Loop).
- Frameworks: Express.js (CRUD Operations, Routing), NestJS (Architectural principles, Modules).
- Concepts: HTTP Requests/Responses, RESTful APIs, Microservices basics (Initial Focus).

Databases & Data Modeling:

- NoSQL: MongoDB, MongoDB Atlas (Cloud deployment), Studio 3T (Local Management).
- Relational (SQL): PostgreSQL, SQL Server, Foundational SQL.
- Data Modeling: Designing and mapping databases using db.diagram.io (ERD and Mapping).

Architecture & Security:

- Architecture: Layered Architecture, Design Patterns (MVC).
- Security: Authentication (Auth) using JWT (JSON Web Tokens), Authorization, Security Middleware (Helmet, Rate Limiting).
- Middleware: Custom Middleware, Validation, Request Logging.

Front-End Basics & Frameworks:

- HTML, CSS, Bootstrap, Tailwind CSS.
- React.js & Next.js (Components, Hooks)

Current Focus:

- Advanced Back-End Design: Deep diving into Microservices and various Design Patterns (Beyond MVC and Layered Architecture).
- Advanced Database Techniques and performance optimization.
- Open-source contributions

PROJECTS & EXPERIENCE

Clinic Management System

- **Tech:** Node.js, Express.js (Layered Architecture), MongoDB Atlas, Mongoose, JWT, Helmet,

Multer, AJV.

- **Overview:** Developed a full-featured Back-End Management System for a psychological clinic, designed to handle complete client and clinic operations, The API is deployed on Vercel.
- **Client Booking:** Implemented a seamless booking process for clients without requiring registration or login.
- **Secure Dashboard:** Designed and built a secure API allowing the doctor to login and fully manage the system.
- **Scheduling:** Handled the scheduling of available appointments (online/offline) and the mechanism for automatic generation of specific booking slots.
- **Data Management:** Enabled management of clinic branches, client files, and session status through complete CRUD operations.
- [\[Live API Demo\]](#) • [\[View Technical\]](#)

ASMA Kitchens (Company Website)

- **Tech:** Next.js, TypeScript, Tailwind CSS, ESLint, React Hooks
- **Overview:** Delivered a modern, performant portfolio website for a kitchen design company (ASMA Kitchens). Built from the ground up using a professional stack with a focus on clean code, responsiveness, and utilizing the Next.js App Router.
[\[Live Demo\]](#) • [\[GitHub Repo\]](#)

My Portfolio

- **Tech:** React, TypeScript, Vite, Tailwind CSS
- **Overview:** This is a modern, responsive personal portfolio website built with React, TypeScript, Vite, and Tailwind CSS. It showcases my projects, skills, and contact information in a clean and interactive way.
[\[Live Demo\]](#) • [\[GitHub Repo\]](#)

Bilingual Search Engine

- **Tech:** Web Scraping, Inverted Indexing, PageRank Algorithm, Database Management
- **Overview:** Co-developed a search engine capable of indexing and searching over 150,000 websites. The engine supports multi-keyword queries in both Arabic and English and uses a ranking algorithm to deliver relevant results.
[\[View Project Details\]](#)

SOFT SKILLS

- Problem Solving • UI/UX Design • Effective Communication
- Teamwork • Time Management • Agile Methodologies

LANGUAGES

- Arabic (Native)
- English (Intermediate – B2)

KEYWORDS (for ATS):

Node.js, Express.js, NestJS, PostgreSQL, MongoDB, JWT, TypeScript, JavaScript, RESTful APIs, SQL Server, Authentication, Authorization, Middleware, Microservices, Layered Architecture, Design Patterns, React.js, Next.js, Git, GitHub, Agile, Testing, Deployment.

